Evaluating the Effects of the Urban Revitalization Plan for Phoenix, Arizona

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Introduction

The concept of a city has come a very long way through history. The first type of permanent settlements in America were colonies, established by multiple groups of people who claimed the land in the new world as theirs. Next, the industrialization era occurred in America, turning cities into economic centers of productivity where people work in factories and mass-produced goods. As the economy of America turned from manufacturing to a service-based economy, people responded by trying to live outside of these urban areas, creating the concept of suburbanization and residential zones. What each phase in history highlights is the fact that over time, cities have rapidly developed and, as a result, contain more complex elements than they ever did before. In this modern era, cities have intricate layouts, which are designed to accommodate all the citizens, boost the economy, and create a culturally and socially tolerable area. Thus, city planning plays a huge role in managing a city and ensuring maximum population, productivity, inclusivity, and leisure for its citizens.

City planners are responsible for managing the city in a way that benefits the economic, political, environmental, and social aspects of people's lives. This is done by enacting policies that alter the landscape such that the investment into policy pays high dividends at the end. In this paper, the focus is turned to a group of policies in a movement called urban revitalization.

What is Urban Revitalization

Urban revitalization (also known as urban regeneration or urban redevelopment) is defined as the means to "impart new life or strength to a neighborhood and revitalize a flagging economy" (Ramlee et al., 2015, p. 361). It is important note that this applies to urban areas only and does not consider the peripherals of the city. Additionally, the policies are placed-based,

which means that a certain policy is applied to specific locations within the urban area. City planners use urban revitalization with many goals in mind. Some of its goals are to “eradicate slums and unhealthy areas”, “create comfortable and safe neighborhoods”, “create better scenery”, and “increase economic in urban areas” (362). The goal of urban revitalization policies is to improve the space of an urban area for social, economic, environmental, and cultural purposes. Urban revitalization is especially vital for city planners to use because “cities tend to age and become obsolete physically and functionally” (362). Thus, planners may also use these policies for maintenance of their cities. Additionally, these policies also aim to address urban challenges that people may have by leveraging resources to enhance “residents’ quality of life and bolster urban competitiveness” (Oh et al., 2024, 1). In essence, the overarching goal of the urban revitalization movement is making the urban area more attractive to citizens and investors, allowing the city to further develop.

Since urban revitalization is a movement rather than a set policy, planners have freedom with how they can utilize land for the purposes of reviving it to boost the urban population. For instance, policies may involve using Business Improved Districts (BID), where “merchants and public owners collective organize to provide public services to improve their established districts” (Sutton, 2008, 42). By creating these, economic productivity is increased in this area of the urban area, which improves the area’s attractiveness to people, thus allowing it to develop into a better area. Alternatively, policies may involve brownfield redevelopment, where areas “which may be contaminated by the presence or potential presence of a hazardous substance, pollutant, or contaminant” is reused (44). Each policy addresses a different aspect of the city, which means the urban revitalization movement can have multiple unique policies that all work in tandem to improve the city in every way possible.

The urban revitalization movement has proven implications that all demonstrate the change in population distribution across a city. Vigdor (2007) conducted a study to measure the effects on the population. He conducted a longitudinal study based on a sample of eighteen metropolitan areas that were going through this movement across two time periods: 1974-1981 and 1985-1993. He then constructed metrics that measured multiple aspects of a good neighborhood and combined them into a composite score for neighborhood quality (pp.2-3). In this study, he found that people value better neighborhood quality, so they are one to three percent more of their annual income for a neighborhood that was one standard deviation higher in neighborhood quality (p.3). This illustrates that people find any improvement in neighborhood quality as a good investment. Additionally, Vigdor (2007) argues that improvements in the quality can contribute to population growth as it increases population capacity and housing supply (p.13). Since people are more willing to pay for their neighborhoods, housing developers have an incentive to build more houses in these urban areas, which is the reason why both housing supply and population can rise. Lastly, Vidgor (2007) also founds that 81% of renters in this area between the two time periods had positively predicted utility changes in revitalizing neighborhoods (p.29). This effect is especially appealing to the population that was in the region before the revitalization plan. Improved quality of life means a higher chance of staying in the region despite higher rents to accommodate for the changes. It also means a higher chance of people moving into the urban areas as people view living in these zones as a worthwhile investment. In conclusion, the study summarized that even though higher costs occur due to the revitalization plan, the overall quality of life increased, and these are just the housing perspective of the plan. There are still cultural, social, and economical aspects of the revitalization plan that can all increase housing and population within the urban areas.

Urban revitalization has proven benefits for communities who need them. However, measuring the policy's effects directly requires data and longitudinal studies like the one conducted by Vigdor. Fortunately, there is a method to predict the long-term effects of policies, which can prove to be very useful when fine-tuning policies to reach their desired outcome. This process involves linear algebra, eigenvalues and eigenvectors more specifically.

Purpose

The goal of this paper is to utilize linear algebra as a tool that planners can use on existing revitalization policies to analyze the plan's efficacy. Linear algebra is the branch of math that deals with vectors and matrices, and its applications are very profound in physics, engineering, machine learning, etc. This paper follows four different applications of linear algebra operations, all of which provide different pieces of information needed to gain a well-rounded prediction of the urban revitalization plan.

Methodology

To do this, a transition matrix will be used, which is a matrix that maps the probability of moving into and out of zones. The reason for using a transition matrix is because it maps out all possibilities of movements between the categorized zones due to the policy, thus getting a full view of how all zones interact with each other and themselves. It is important to note that the paper simplifies the city into four distinct zones to illustrate the effects in a simplified and intuitive manner. There can be many more zones, and this transition matrix will be bigger or smaller. For the purposes of this project, the transition matrix that is used is:

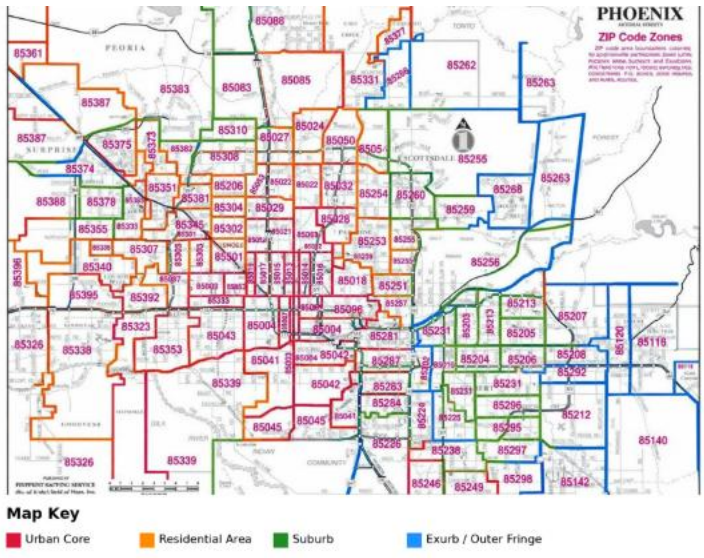
$$PH = \begin{bmatrix} 0.70 & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60 & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70 & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75 \end{bmatrix}$$

Row and column 1 correspond with the Core (downtown), 2 with the Inner Ring (Urban Residential), 3 with the Suburbs, and 4 with the exurbs. Lastly, each entry PH_{ij} is the probability of moving from zone j to zone i. All values will be rounded to the nearest thousandth.

As mentioned before, there are four tasks that will be done with this matrix. The first task is to prove that the matrix has an eigenvalue of 1, to prove that there is an equilibrium population distribution. Next, a steady state vector will be found for the matrix, which is a vector that resembles the final population after many years. Thirdly, a sensitivity analysis will be conducted, and a steady state vector will be recalculated, allowing for comparison between the populations of each zone due to investment. Lastly, the second highest eigenvalue will be determined, which provides insight as to how fast the population reaches an equilibrium distribution.

Project Background Information

Phoenix is the capital of Arizona. It has a population of 1,673,164 people, and it has a size of 518 square miles, hence having a population density of 3104.5 people per square mile. It is in South Central Arizona, in the Salt River Valley, and it is surrounded by mountain ranges and the Sonoran Desert. As shown in the image on the left, there are 74 zip codes in Phoenix. 15 zip codes make up the core, 22 make up the residential area (inner ring), 23 make up the suburbs, and 14 make up the exurbs. The area consists of an equal number of males and females, with most of the population consisting of white and Hispanic people. Economically, the median household income is $81,332 and the median per capita income is $41,981. Phoenix is currently going through a revitalization plan, called the RIO PHX. The RIO PHX project plans to transform the 20-mile stretch of the Salt River that runs through Phoenix into a more socially and economically active zone, essentially turning it into a part of downtown Phoenix. They plan to open investment along the corridor, create housing and job opportunities, add parks and recreation areas, restore native ecosystems and habitats, and improve the health of nearby communities and make Phoenix more walkable. This initiative was launched in 2023, and development plans are in progress today. This project is modelled based on RIO PHX, and the math that is done is to predict the effect that this revitalization plan will have on all zones of Phoenix.



Proving that an equilibrium population distribution exists

An equilibrium population distribution is a population distribution that remains unchanged after the transition matrix is applied. For there to be an equilibrium population distribution, there must exist an eigenvalue of 1, since $Ax = \lambda x$ simplifies to $Ax = x$, where A is the transition matrix, x is the population distribution, and λ is the eigenvalue.

$$PH = \begin{bmatrix} 0.70 & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60 & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70 & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75 \end{bmatrix}$$

Eigenvalues are found by solving the equation $\det(PH - \lambda I) = 0$, where I represents the identity matrix. To prove $\lambda = 1$ exists, there must be a solution for $\det(PH-I) = 0$. First, the matrix PH-I must be constructed

$$PH - I = \begin{bmatrix} 0.70 & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60 & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70 & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.30 & 0.15 & 0.05 & 0.10 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.30 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix}$$

It is important to note that manipulating one row into a row of all 0s through row operations implies that the determinant of the matrix is 0, since cofactor expansion along that row would result in an answer of 0. Also, adding scalar multiples of rows to one row of the matrix does not change the determinant of the original matrix. So, by adding row 2 to row 1, row 3 to row 1, and row 4 to row 1, the first row becomes a row of all 0s:

$$\begin{bmatrix} -0.30 & 0.15 & 0.05 & 0.10 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.30 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix} \xrightarrow{R1+R2 \to R1} \begin{bmatrix} -0.20 & -0.25 & 0.25 & 0.15 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.30 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix} \xrightarrow{R1+R3 \to R1}$$

$$\begin{bmatrix} -0.10 & -0.05 & -0.05 & 0.25 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.30 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix} \xrightarrow{R1+R4 \to R1} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.30 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix}$$

Hence, this implies det (PH-I) = 0, which also proves that det (PH – λI) = 0 when λ = 1. And since an eigenvalue of 1 proves the equation Ax = x, and hence there exists an equilibrium population.

Finding the steady state vector for the population:

The steady state vector of a transition matrix represents the population distribution of a city after a long time. The vector resembles the what the population distribution to settle to after many, many transitions between zones, hence using trends to give each zone a unique number of people that are expected after a long time. Hence, this is especially important to predict the population distribution in the future, and to evaluate the long-term effects of an urban plan on the original population as they transition between zones for years.

$$\text{PH-I} = \begin{bmatrix} -0.30 & 0.15 & 0.05 & 0.10 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.30 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix}$$

To find the steady state vector, there must be a solution for the equation (PH- I)x = 0 where x = $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ is the steady state vector. First, the matrix PH – I is transformed into RREF (reduced row echelon form). This is done so that the matrix has 1s on the main diagonal and 0s elsewhere, providing a solution for the matrix. RREF requires three fundamental row operations that do not change the information carried by a matrix: switching rows, multiplying rows by a scalar, and

adding scalar multiples of rows to other rows. The operations listed below outline one of the many ways the matrix is converted to RREF:

$$\begin{bmatrix} -0.30 & 0.15 & 0.05 & 0.10 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.30 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix} \xrightarrow{R1 \leftrightarrow R2} \begin{bmatrix} 0.10 & -0.40 & 0.20 & 0.05 \\ -0.30 & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.20 & -0.30 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix} \xrightarrow[-R1+R4 \to R4]{3R1+R2 \to R2 \\ -R1+R3 \to R3}$$

$$\begin{bmatrix} 0.10 & -0.40 & 0.20 & 0.05 \\ 0 & -1.05 & 0.65 & 0.25 \\ 0 & 0.60 & -0.50 & 0.05 \\ 0 & 0.45 & -0.15 & -0.30 \end{bmatrix} \xrightarrow[\frac{R2}{-1.05}]{\frac{R1}{0.10}} \begin{bmatrix} 1 & -4 & 2 & 0.5 \\ 0 & 1 & -0.619 & -0.238 \\ 0 & 0.60 & -0.50 & 0.05 \\ 0 & 0.45 & -0.15 & -0.30 \end{bmatrix} \xrightarrow[-0.45R2+R4 \to R4]{4R2+R1 \to R1 \\ -0.6R2+R3 \to R3}$$

$$\begin{bmatrix} 1 & 0 & -0.476 & -0.452 \\ 0 & 1 & -0.619 & -0.238 \\ 0 & 0 & -0.129 & 0.193 \\ 0 & 0 & 0.129 & -0.193 \end{bmatrix} \xrightarrow{R3+R4 \to R4} \begin{bmatrix} 1 & 0 & -0.476 & -0.452 \\ 0 & 1 & -0.619 & -0.238 \\ 0 & 0 & -0.129 & 0.193 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{R3}{-0.129}}$$

$$\begin{bmatrix} 1 & 0 & -0.476 & -0.452 \\ 0 & 1 & -0.619 & -0.238 \\ 0 & 0 & 1 & -1.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[0.476R3+R1 \to R1]{0.169R3+R2 \to R2} \begin{bmatrix} 1 & 0 & 0 & -1.167 \\ 0 & 1 & 0 & -1.167 \\ 0 & 0 & 1 & -1.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Now that the matrix is converted, this equation can be solved much easier. Column 1 represents x_1, column 2 represents x_2, and so on. Since the fourth row is a row of all 0s, there is a free variable, which is a variable that can be any real number. This also means the matrix has infinite solutions. Since the last column is the only column that does not have a leading 1, x_4 is the free variable. To find the general solution, each variable of the vector can be parameterized. By letting $x_4 = t$, the steady state vector can be derived by solving the equation:

$$x_1 - 1.167x_4 = 0 \longrightarrow x_1 = 1.167t$$

$$x_2 - 1.167x_4 = 0 \longrightarrow x_2 = 1.167t$$

$$x_3 - 1.5x_4 = 0 \longrightarrow x_3 = 1.5t$$

The steady state vector is x = $\begin{bmatrix} 1.167t \\ 1.167t \\ 1.5t \\ t \end{bmatrix}$.

Since the population distribution must add up to 1, the steady state vector needs to be normalized to provide the information necessary for analysis. To normalize is a simple process that requires setting the sum of all entries in the vector to 1 and solving for t. Thus,

$$1.167t + 1.167t + 1.5t + t = 1$$

$$4.834t = 1$$

$$t = \frac{1}{4.834}$$

Hence, x = $\begin{bmatrix} 1.167(\frac{1}{4.834}) \\ 1.167(\frac{1}{4.834}) \\ 1.5(\frac{1}{4.834}) \\ \frac{1}{4.834} \end{bmatrix} = \begin{bmatrix} 0.241 \\ 0.241 \\ 0.310 \\ 0.207 \end{bmatrix}$

The analysis of the population distribution can now be done. Due to the plan, 24.1% of Phoenix residents will live in the core, 24.1% will live in the Inner Ring, 31% will live in the suburbs, and 20.7% will live in the exurbs over the course of many years in the future.

Sensitivity Analysis on Investment for a New Rail System

The city of Phoenix has decided to invest $500 million in a new light rail system. Hence, this changes the transition matrix as improvement allows for better accessibility to other zones. Hence the probability of moving from the Suburbs to the Core is changed from 0.05 to 0.15. On the other hand, the probability of staying in suburbs changes from 0.7 to 0.6, since increasing the probability of moving decreases the probability of staying. Hence, the new transition matrix is:

$$PH = \begin{bmatrix} 0.70 & 0.15 & 0.15 & 0.10 \\ 0.10 & 0.60 & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.60 & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75 \end{bmatrix}$$

To determine the effects of this rail system on the population, another steady state vector PH-I will be determined, but PH represents the new transition matrix. Then, a comparison of the two steady state vectors will be made to determine the number of people who relocated after a policy is enacted in the long run.

$$PH - I = \begin{bmatrix} 0.70 & 0.15 & 0.15 & 0.10 \\ 0.10 & 0.60 & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.60 & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75 \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -0.30 & 0.15 & 0.15 & 0.10 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.40 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix}$$

Now the matrix is converted to RREF, doing elementary row operations again in the exact same manner as the previous one:

$$\begin{bmatrix} -0.30 & 0.15 & 0.15 & 0.10 \\ 0.10 & -0.40 & 0.20 & 0.05 \\ 0.10 & 0.20 & -0.40 & 0.10 \\ 0.10 & 0.05 & 0.05 & -0.25 \end{bmatrix} \xrightarrow{R1 \leftrightarrow R2} \begin{bmatrix} 0.10 & -0.40 & 0.20 & 0.05 & 0 \\ -0.30 & 0.15 & 0.15 & 0.10 & 0 \\ 0.10 & 0.20 & -0.40 & 0.10 & 0 \\ 0.10 & 0.05 & 0.05 & -0.25 & 0 \end{bmatrix} \xrightarrow[-R1+R4 \to R4]{\substack{3R1+R2 \to R2 \\ -R1+R3 \to R3}}$$

$$\begin{bmatrix} 0.10 & -0.40 & 0.20 & 0.05 & 0 \\ 0 & -1.05 & 0.75 & 0.25 & 0 \\ 0 & 0.60 & -0.60 & 0.05 & 0 \\ 0 & 0.45 & -0.15 & -0.30 & 0 \end{bmatrix} \xrightarrow[-1.05]{\frac{R1}{0.10}\ R2} \begin{bmatrix} 1 & -4 & 2 & 0.5 & 0 \\ 0 & -1.05 & 0.75 & 0.25 & 0 \\ 0 & 0.60 & -0.60 & 0.05 & 0 \\ 0 & 0.45 & -0.15 & -0.30 & 0 \end{bmatrix} \xrightarrow[-0.45R2+R4\to R4]{4R2+R1\to R1 \\ -0.6R2+R3\to R3}$$

$$\begin{bmatrix} 1 & 0 & -0.857 & -0.452 & 0 \\ 0 & 1 & -0.714 & -0.238 & 0 \\ 0 & 0 & -0.171 & 0.193 & 0 \\ 0 & 0 & 0.171 & -0.193 & 0 \end{bmatrix} \xrightarrow{R3+R4\to R4} \begin{bmatrix} 1 & 0 & -0.857 & -0.452 & 0 \\ 0 & 1 & -0.714 & -0.238 & 0 \\ 0 & 0 & -0.171 & 0.193 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[-0.171]{\frac{R3}{}}$$

$$\begin{bmatrix} 1 & 0 & -0.857 & -0.452 & 0 \\ 0 & 1 & -0.714 & -0.238 & 0 \\ 0 & 0 & 1 & -1.125 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow[0.857R3+R1\to R1]{0.714R3+R2\to R2} \begin{bmatrix} 1 & 0 & 0 & -1.417 & 0 \\ 0 & 1 & 0 & -1.042 & 0 \\ 0 & 0 & 1 & -1.125 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Once again, there is a free variable. By letting x_4= t:

$$x_1 - 1.417x_4 = 0 \longrightarrow x_1 = 1.417t$$

$$x_2 - 1.042x_4 = 0 \longrightarrow x_2 = 1.042t$$

$$x_3 - 1.125x_4 = 0 \longrightarrow x_3 = 1.125t$$

So, $x_c = \begin{bmatrix} 1.417t \\ 1.042t \\ 1.125t \\ t \end{bmatrix}$.

Once again, the vector needs to be normalized:

$$1.417t + 1.042t + 1.125t + t = 1$$

$$4.584t = 1$$

$$t = \frac{1}{4.584}$$

Hence, $x_c = \begin{bmatrix} 1.417(\frac{1}{4.584}) \\ 1.042(\frac{1}{4.584}) \\ 1.125(\frac{1}{4.584}) \\ \frac{1}{4.584} \end{bmatrix} = \begin{bmatrix} 0.309 \\ 0.227 \\ 0.245 \\ 0.218 \end{bmatrix}$

By multiplying both vectors to the population of Phoenix, the number of people per zone can be determined before and after the policy (rounding to the nearest whole number):

$$x = 1673164 \begin{bmatrix} 0.241 \\ 0.241 \\ 0.310 \\ 0.207 \end{bmatrix} = \begin{bmatrix} 403233 \\ 403233 \\ 518681 \\ 346345 \end{bmatrix}$$

$$x_c = 1673164 \begin{bmatrix} 0.309 \\ 0.227 \\ 0.245 \\ 0.218 \end{bmatrix} = \begin{bmatrix} 517008 \\ 379808 \\ 409925 \\ 364750 \end{bmatrix}$$

With these populations, the sensitivity analysis due to a policy can be determined. For example, the new system has resulted in 113,775 people relocating to the Core from the other three zones.

Determining the Rate of Convergence for Eigenvalues:

Each eigenvalue determines the speed at which a population distribution approaches a steady state. The goal of this task is to find the second largest eigenvalue. To determine the second largest eigenvalue, all eigenvalues must be found by solving the equation det (PH-λI) = 0.

$$PH - \lambda I = \begin{bmatrix} 0.70 & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60 & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70 & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 0 \\ 0 & 0 & 0 & \lambda \end{bmatrix}$$ (scalar multiplication to identity)

$$PH\text{-}\lambda I = \begin{bmatrix} 0.70 - \lambda & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60 - \lambda & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70 - \lambda & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75 - \lambda \end{bmatrix}$$

$$\det(PH-\lambda I) = \begin{vmatrix} 0.70-\lambda & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60-\lambda & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70-\lambda & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75-\lambda \end{vmatrix}$$

To simplify the process, R1, R2, and R3 are added to R4 (does not change determinant) to get a row with equal values:

$$\begin{vmatrix} 0.70-\lambda & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60-\lambda & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70-\lambda & 0.10 \\ 0.10 & 0.05 & 0.05 & 0.75-\lambda \end{vmatrix} \xrightarrow{R4+(R1+R2+R3)\rightarrow R4} \begin{vmatrix} 0.70-\lambda & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60-\lambda & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70-\lambda & 0.10 \\ 1-\lambda & 1-\lambda & 1-\lambda & 1-\lambda \end{vmatrix}$$

Factoring out a $1-\lambda$ from the determinant results with:

$$\begin{vmatrix} 0.70-\lambda & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60-\lambda & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70-\lambda & 0.10 \\ 1-\lambda & 1-\lambda & 1-\lambda & 1-\lambda \end{vmatrix} = (1-\lambda) \begin{vmatrix} 0.70-\lambda & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60-\lambda & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70-\lambda & 0.10 \\ 1 & 1 & 1 & 1 \end{vmatrix}$$

To further simplify, columns C2, C3, and C4 are subtracted to C1 to get the last row as [1,0,0,0]:

$$(1-\lambda) \begin{vmatrix} 0.70-\lambda & 0.15 & 0.05 & 0.10 \\ 0.10 & 0.60-\lambda & 0.20 & 0.05 \\ 0.10 & 0.20 & 0.70-\lambda & 0.10 \\ 1 & 1 & 1 & 1 \end{vmatrix} \xrightarrow[C4-C1\rightarrow C4]{\substack{C2-C1\rightarrow C2 \\ C3-C1\rightarrow C3}}$$

$$(1-\lambda) \begin{vmatrix} 0.70-\lambda & \lambda-0.55 & \lambda-0.65 & \lambda-0.60 \\ 0.10 & 0.50-\lambda & 0.10 & -0.05 \\ 0.10 & 0.10 & 0.60-\lambda & 0 \\ 1 & 0 & 0 & 0 \end{vmatrix}$$

The determinant can be found through cofactor expansion using the last row. The first term in the expansion is negative ($(-1)^{4+1} = -1$), and the other terms are 0 (0*any determinant = 0), so:

$$-1(1-\lambda) \begin{vmatrix} \lambda-0.55 & \lambda-0.65 & \lambda-0.60 \\ 0.50-\lambda & 0.10 & -0.05 \\ 0.10 & 0.60-\lambda & 0 \end{vmatrix}$$

Cofactor expansion is conducted on the last row of this 3x3 determinant. The first term in the expansion is positive (((-1)$^{3+1}$ = 1), and then alternates:

$$\begin{vmatrix} \lambda - 0.55 & \lambda - 0.65 & \lambda - 0.60 \\ 0.50 - \lambda & 0.10 & -0.05 \\ 0.10 & 0.60 - \lambda & 0 \end{vmatrix} =$$

$$0.10 \begin{vmatrix} \lambda - 0.65 & \lambda - 0.60 \\ 0.10 & -0.05 \end{vmatrix} - (0.60 - \lambda) \begin{vmatrix} \lambda - 0.55 & \lambda - 0.60 \\ 0.50 - \lambda & -0.05 \end{vmatrix} + 0$$

Using the fact that $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$, the 3x3 determinant is simplified to:

$$0.10[(-0.05)(\lambda - 0.65) - (0.10)(\lambda - 0.60)] - (0.60 - \lambda)[(\lambda - 0.55)(-0.05) - (0.50 - \lambda)(\lambda - 0.60)$$

$$= \lambda^3 - 1.75\lambda^2 + 1.0025\lambda - 0.18725$$

Substituting back into the rest of the terms results in a characteristic equation:

$$-1(1 - \lambda)(\lambda^3 - 1.75\lambda^2 + 1.0025\lambda - 0.18725)$$

$$= \lambda^4 - 2.75\lambda^3 - 2.7525\lambda^2 - 1.18975\lambda + 0.18725$$

The solution needs to be solved, which can be reasonably approximated using a graphing device. The eigenvalues, approximated to the nearest thousandth, are:

$$\lambda_1 = 1, \lambda_2 = 0.700, \lambda_3 = 0.615, \lambda_4 = 0.435$$

The second largest eigenvalue is 0.7.

A further note, since there are four unique eigenvalues, there are four unique eigenvectors that are linearly independent of each other (cannot combine the four vectors to create the four vectors). This means any 4x1 vector (like the steady state vector) can be created by this equation:

$x = C_1V_1 + C_2V_2 + C_3V_3 + C_4V_4$ where C represents a constant and V is an eigenvector. Using this, another equation can be created that lists the steady state vector after a specific number of transitions:

$$(PH)^n x = C_1\lambda_1{}^n V_1 + C_2\lambda_2{}^n V_2 + C_3\lambda_3{}^n V_3 + C_4\lambda_4{}^n V_4$$

This is done by left hand multiplication, and the fact that multiplying a matrix to an eigenvector is the same as multiplying its eigenvalue to its eigenvector. N represents the number of transitions, and x represents the initial population vector. Substituting the eigenvalues into the equation simplifies the right-hand side of the equation to this:

$$C_1(1)^n V_1 + C_2(0.7)^n V_2 + C_3(0.615)^n V_3 + C_4(0.435)^n V_1$$

This equation is significant in figuring how many years it will take for a population to reach its steady state, which is the population in the long run. As n increases, the terms slowly decrease approaching 0 since the values that are raised are less than 1. This also means that the fourth term will decrease the fastest, followed by the third, and lastly the second term. This is the significance of the second largest eigenvalue; it dictates the speed at which the population converges into a stable population distribution. A higher value means that the population reaches equilibrium slowly, while a smaller eigenvalue means that the population reaches equilibrium quickly.

To recap, four different applications of linear algebra were demonstrated to quantify the effects of urban revitalization. The first task provided insight as to whether the population would approach a stable equilibrium. The second task quantified the long-term effects in terms of population distribution. The third task allowed for the tracking of changes caused by enactments

of different policies. And lastly, the fourth task provided insight as to the speed at which this population reaches stability.

Conclusion

It is clear from the scenario of RIO PHX that urban revitalization policies have a big impact on the population distribution of a city. These changes are easily mapped out computationally using linear algebra. Each task provided information that a policy maker could use to determine if their policies have the desired outcome. These are four of many different applications of linear algebra that can be used for a humanities field like urban policies.

There is an obvious limitation in predictions as there are numerous factors that affect people's motives towards moving places that are beyond the benefits of urban revitalization. However, this study can be repeated with bigger matrices to account for all these variables. So, linear algebra remains an accurate way to measure the efficacy of a policy.

Another benefit of linear algebra remains is the ability to develop code. The math in this paper is written out, but it would be faster to code. Python libraries like NumPy make computing steady state vectors and determinants faster, which allows for the introduction of bigger and more complex matrices.

In the end, the reason for why linear algebra should be implemented into these urban revitalization policies is for the citizens. People are willing to spend extra to get a better quality of life, and mapping out the exact effects of the plan will ensure all citizens can achieve that.

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